Lemma A.3. *Let $B_\delta(\theta)$ denote the open ball of radius δ centered at $\theta \in \Theta := \mathbb{R} \setminus \{0\} \times \mathbb{R}^2_+$. For any $S \in \mathcal{S}$ and $\theta \in \Theta$, let $\mathcal{D}_{obs} = (X_i)_{1 \le i \le n}$ be n i.i.d. observations generated by (1) with law $P_{m(S,\theta)}$ and density $f(x \mid S, \theta)$. The following four conditions hold*

(i) for all $x_1^n = (x_i)_{1\le i \le n}$ with $x_i \in \mathbb{R}^2$ and $\theta \in \Theta$, $\prod_{i=1}^n f(x_i \mid S, \theta) > 0$. Moreover, for all x_1^n, $\ell_n(\theta \mid S) = \sum_{i=1}^n \log f(x_i \mid S, \theta)$ is six times continuously differentiable in θ;

(ii) for any $\theta_0 \in \Theta$ there exist $\epsilon > 0$ and $M < \infty$ such that $B_\epsilon(\theta_0) \subseteq \Theta$ and for $1 \le j_1, \ldots, j_d \le m$ with $d \le 6$,

$$\limsup_{n\to\infty} \sup\{n^{-1} |\partial_{j_1} \ldots \partial_{j_d} \ell_n(\theta \mid S)| : \theta \in B_\epsilon(\theta_0)\} < M,$$

with $P_{m(S,\theta_0)}$-probability one;

(iii) for any $\theta_0 \in \Theta$ and some $\epsilon > 0$,

$$\limsup_{n\to\infty} \sup\{\det(n^{-1} \nabla^2_\theta \ell_n(\theta \mid S)) : \theta \in B_\epsilon(\theta_0)\} < 0$$

with $P_{m(S,\theta_0)}$-probability one;

(iv) for any $\theta_0 \in \Theta$ and any $\delta > 0$,

$$\limsup_{n\to\infty} \sup\{n^{-1}[\ell_n(\theta \mid S) - \ell_n(\theta_0 \mid S)] : \theta \in \Theta \setminus B_\delta(\theta_0)\} < 0$$

with $P_{m(S,\theta_0)}$-probability one.

Hence, the models are Laplace regular as per Kass et al. [1990].

Proof. We check the four conditions presented in Kass et al. [1990] which allow us to approximate the marginal likelihoods using Laplace's method. Without loss of generality we assume $S = S^1$. The other remaining cases follow along the same lines. Under S^1, the log-likelihood $\ell_n(\theta \mid S^1)$ as a function of the MLE $\hat{\theta}^1$ is

$$\ell_n(\theta \mid S^1) = \log = -n \log 2\pi - \frac{n}{2} \log \tau_1^2 \tau_2^2 - \frac{n}{2} \cdot \frac{(w - \hat{w})^2 \hat{\tau}_2^2}{\tau_1^2} - \frac{n}{2} \cdot \frac{\hat{\tau}_1^2}{\tau_1^2} - \frac{n}{2} \cdot \frac{\hat{\tau}_2^2}{\tau_2^2}. \tag{19}$$

(i) We note that $\ell_n(\theta \mid S^1)$ is 6-times continuously differentiable. Moreover, since the observations are assumed to be independent, the joint density function for x_1^n is $\prod_{i=1}^n f(x_i \mid S, \theta)$ which is non-negative because each term in the product is strictly positive,

$$f(x_i \mid S^1, \theta) = \frac{1}{\sqrt{2\pi\tau_1^2}} \exp\left\{-\frac{(x_i(1) - w x_i(2))^2}{2\tau_1^2}\right\} \frac{1}{\sqrt{2\pi\tau_2^2}} \exp\left\{-\frac{x_i(2)^2}{2\tau_2^2}\right\} > 0.$$

(ii) The log-likelihood function $\ell_n(\theta \mid S_1)$ decomposes into a sum of terms that are either quadratic polynomials in the structural parameter w or rational functions of the variance parameters τ_1^2, τ_2^2.

Specifically, partial derivatives with respect to w of order $d \ge 3$ vanish and hence are bounded. Partial derivatives with respect to the variances involve terms of the form τ^{-k}, which are C^∞ smooth and bounded on any compact domain where the variances are bounded away from zero (i.e., $\tau_j^2 \ge \delta > 0$). Consequently, all mixed partial derivatives of order $d \ge 3$, when normalized by n^{-1}, are uniformly bounded in a neighborhood of the true parameter, satisfying the necessary condition.

(iii) The Hessian is

$$H(\theta) := H(w, \tau_1^2, \tau_2^2) = \begin{bmatrix} -\frac{n\hat{\tau}_2^2}{\tau_1^2} & \frac{n\hat{\tau}_2^2(w-\hat{w})}{(\tau_1^2)^2} & 0 \\ \frac{n\hat{\tau}_2^2(w-\hat{w})}{(\tau_1^2)^2} & \frac{n}{2(\tau_1^2)^2} - \frac{n}{(\tau_1^2)^3}\left[\hat{\tau}_1^2 + \hat{\tau}_2^2(w-\hat{w})^2\right] & 0 \\ 0 & 0 & \frac{n}{2(\tau_2^2)^2} - \frac{n\hat{\tau}_2^2}{(\tau_2^2)^3} \end{bmatrix} \tag{20}$$

with the determinant of the normalized Hessian being

$$h_n(\theta) := \det\left(\frac{1}{n}H(\theta)\right) = -\frac{\hat{\tau}_2^2}{4\tau_1^6\tau_2^4}\Big(1 - \frac{2\hat{\tau}_2^2}{\tau_2^2}\Big)\Big(1 - \frac{2\hat{\tau}_1^2}{\tau_1^2}\Big). \tag{21}$$

Let $\theta_0 := \big(w_0, \tau_{1,0}^2, \tau_{2,0}^2\big) \in \Theta$ be the true generating parameter and $d := \min\{\tau_{1,0}^2, \tau_{2,0}^2\} > 0$. Consider the closed ball $\bar{B}_{\epsilon_1}(\theta)$ with $\epsilon_1 = d/2$. Since both variances are bounded away from 0, $h_n(\theta)$ is continuous and well-defined on $\bar{B}_{\epsilon_1}(\theta_0)$. Furthermore, both estimators $\hat{\tau_1}^2$ and $\hat{\tau_2}^2$ are consistent per (13) and Lemma A.1,

$$\hat{\tau}_1^2 \xrightarrow{a.s.} \tau_{1,0}^2 \quad \text{and} \quad \hat{\tau}_2^2 \xrightarrow{a.s.} \tau_{2,0}^2,$$

because $\mathbb{E}[X(1)^2] = w_0^2\tau_{2,0}^2 + \tau_{1,0}^2$ and $\mathbb{E}[X(2)^2] = \tau_{2,0}^2$. Therefore, because $h_n(\theta)$ is a continuous function of the estimators, by the continuous mapping theorem, it converges uniformly and almost surely over $\bar{B}_{\epsilon_1}(\theta_0)$

$$h_n(\theta) \xrightarrow{a.s.} h_\infty(\theta) := -\frac{\tau_{2,0}^2}{4\tau_1^6\tau_2^4}\left(1 - \frac{2\tau_{2,0}^2}{\tau_2^2}\right)\left(1 - \frac{2\tau_{1,0}^2}{\tau_1^2}\right),$$

with $h_\infty(\theta_0) = -1/(4\tau_{1,0}^6\tau_{2,0}^2) =: -C < 0$. Since $h_\infty(\theta)$ is continuous on $\bar{B}_{\epsilon_1}(\theta_0)$, there exist $\epsilon^* < \epsilon_1$ such that

$$h_\infty(\theta) \le -\frac{C}{2} < 0, \text{ for all } \theta \in B_{\epsilon^*}(\theta_0),$$

where $B_{\epsilon^*}(\theta_0)$ is an open ball centered at θ_0. Since uniform convergence on $\bar{B}_{\epsilon_1}(\theta_0)$ implies uniform convergence on its subset $B_{\epsilon^*}(\theta_0)$, we can pass the limit through the supremum

$$\lim_{n\to\infty} \sup_{\theta\in B_{\epsilon^*}(\theta_0)} h_n(\theta) = \sup_{\theta\in B_{\epsilon^*}(\theta_0)} h_\infty(\theta) \le -\frac{C}{2} < 0 \quad \text{a.s.}$$

Because this limit exists and is strictly negative, the limit superior is also strictly negative, yielding the desired result.

(iv) Condition (iv) requires the global consistency of the MLE. For simplicity, we consider only the S^1 case; the other cases follow analogously. Let $\hat{\theta}^1 = [\hat{w}, \hat{\tau}_1^2, \hat{\tau}_2^2]^\top$ denote the MLE under model S^1. Using the expansion in (19), the normalized difference between the log-likelihood at a parameter $\theta = [w, \tau_1^2, \tau_2^2]^\top$ and at the MLE is given by

$$g_n(\theta) := \frac{1}{n}\Big[\ell_n(\theta \mid S^1) - \ell_n(\hat{\theta}^1 \mid S^1)\Big] = -\frac{1}{2}\left(\frac{(w-\hat{w})^2\hat{\tau}_2^2}{\tau_1^2} + \phi\left(\frac{\hat{\tau}_1^2}{\tau_1^2}\right) + \phi\left(\frac{\hat{\tau}_2^2}{\tau_2^2}\right)\right), \tag{22}$$

where we define $\phi(u) = u - 1 - \log u$ for $u > 0$. It is a standard result that $\phi(u) \ge 0$ for all $u > 0$, with equality holding if and only if $u = 1$. Furthermore, $\phi(u)$ is strictly decreasing on $(0,1)$, strictly increasing on $(1,\infty)$, and satisfies $\lim_{u\to 0}\phi(u) = \lim_{u\to\infty}\phi(u) = \infty$. Because each term inside the bracket is non-negative, $g_n(\theta) \le 0$ globally, with equality strictly at $\theta = \hat{\theta}^1$.

The primary technical challenge is that the parameter space $\Theta = \mathbb{R}\setminus\{0\}\times(0,\infty)^2$ is not compact. We therefore proceed by partitioning $\Theta \setminus B_\delta(\theta_0)$ into a non-compact region and a compact region.

By Lemma A.2, the MLE is consistent, i.e. $\hat{\theta}^1 \xrightarrow{a.s.} \theta_0$. Consequently, the sequence $\hat{\theta}^1$ is eventually bounded almost surely. There exist positive constants c, C such that for n sufficiently large, we have $\hat{\tau}_1^2, \hat{\tau}_2^2 \in [c, C]$ and $|\hat{w}| \le C$.

Fix an arbitrary constant $M > 0$. We construct a compact set $K := [-W, W] \times [l, L] \times [l, L]$ containing $B_\delta(\theta_0)$ by choosing the positive constants W, l, L such that

$$L > C \quad \text{and} \quad \phi\left(\frac{C}{L}\right) > 2M \tag{23}$$

$$l < c \quad \text{and} \quad \phi\left(\frac{c}{l}\right) > 2M \tag{24}$$

$$W > C + \sqrt{\frac{2ML}{c}} \tag{25}$$

Conditions (23) and (24) are trivially satisfied by taking L sufficiently large and l sufficiently small, due to the behavior of ϕ when $u \to 0$ and $u \to \infty$. Next, we demonstrate that for any n large enough and any $\theta \in \Theta \setminus K$, we have $g_n(\theta) < -M$. We evaluate the following three cases

– **Case 1:** If $\tau_i^2 > L$ for at least one $i \in \{1,2\}$, then $\frac{\hat{\tau}_i^2}{\tau_i^2} < \frac{C}{L} < 1$. Because ϕ is strictly decreasing on $(0,1)$, and the condition in (23) implies $\phi\left(\frac{\hat{\tau}_i^2}{\tau_i^2}\right) > \phi\left(\frac{C}{L}\right) > 2M$. This ensures $g_n(\theta) < -M$.

– **Case 2:** If $\tau_i^2 < l$ for at least one $i \in \{1,2\}$, then $\frac{\hat{\tau}_i^2}{\tau_i^2} > \frac{c}{l} > 1$. Because ϕ is strictly increasing on $(1,\infty)$, and the condition in (24) implies $\phi\left(\frac{\hat{\tau}_i^2}{\tau_i^2}\right) > \phi\left(\frac{c}{l}\right) > 2M$. This ensures $g_n(\theta) < -M$.

– **Case 3:** Assume θ is not caught by Cases 1 or 2, which guarantees $\tau_1^2 \leq L$. Because $\theta \notin K$, it must be that $|w| > W$. We then have $\frac{(w-\hat{w})^2\hat{\tau}_2^2}{\tau_1^2} \geq \frac{(|w|-|\hat{w}|)^2 c}{L} \geq \frac{(W-C)^2 c}{L}$. By the condition in (25), this term is strictly greater than $2M$, ensuring $g_n(\theta) < -M$.

Thus, we have established that $\sup_{\theta\in\Theta\setminus K} g_n(\theta) \leq -M$ eventually almost surely.

We now restrict our attention to the compact set, $K \setminus B_\delta(\theta_0)$, bounded away from the true parameter. Because $\hat{\theta}^1 \xrightarrow{a.s.} \theta_0$ and $g_n(\theta)$ is continuous in both θ and $\hat{\theta}^1$, the continuous mapping theorem yields pointwise convergence

$$g_n(\theta) \xrightarrow{a.s.} g_\infty(\theta) = -\frac{1}{2}\left(\frac{(w-w_0)^2\tau_{2,0}^2}{\tau_1^2} + \phi\left(\frac{\tau_{1,0}^2}{\tau_1^2}\right) + \phi\left(\frac{\tau_{2,0}^2}{\tau_2^2}\right)\right).$$

Moreover, since $K \setminus B_\delta(\theta_0)$ is compact, the above convergence implies uniform converge as well, i.e.

$$\sup_{\theta\in K\setminus B_\delta(\theta_0)} |g_n(\theta) - g_\infty(\theta)| \xrightarrow{a.s.} 0.$$

Notice that $g_\infty(\theta) \leq 0$ everywhere, with $g_\infty(\theta) = 0$ if and only if $\theta = \theta_0$. Because $\theta_0 \notin K \setminus B_\delta(\theta_0)$ and the set is compact, $g_\infty(\theta)$ must attain a strictly negative maximum on this set. Let $\sup_{\theta\in K\setminus B_\delta(\theta_0)} g_\infty(\theta) = -M'$ with $M' > 0$. Therefore

$$\begin{aligned}
\sup_{\theta\in K\setminus B_\delta(\theta_0)} g_n(\theta) &= \sup_{\theta\in K\setminus B_\delta(\theta_0)} \left(g_n(\theta) - g_\infty(\theta) + g_\infty(\theta)\right) \\
&\leq \sup_{\theta\in K\setminus B_\delta(\theta_0)} g_\infty(\theta) + |g_n(\theta) - g_\infty(\theta)| \\
&\leq \sup_{\theta\in K\setminus B_\delta(\theta_0)} g_\infty(\theta) + \sup_{\theta\in K\setminus B_\delta(\theta_0)} |g_n(\theta) - g_\infty(\theta)| \leq -M' + \frac{M'}{2} = -\frac{M'}{2},
\end{aligned}$$

eventually almost surely. Combining the bounds from the two regions, we bound the supremum over the entire set $\Theta \setminus B_\delta(\theta_0)$

$$\sup_{\theta\in\Theta\setminus B_\delta(\theta_0)} g_n(\theta) = \max\left\{\sup_{\theta\in K\setminus B_\delta(\theta_0)} g_n(\theta),\ \sup_{\theta\in\Theta\setminus K} g_n(\theta)\right\} \leq \max\{-M'/2, -M\} < 0.$$

Taking the limit supremum yields

$$\limsup_{n\to\infty} \sup_{\theta\in\Theta\setminus B_\delta(\theta_0)} g_n(\theta) < 0 \quad \text{a.s.}$$

Finally, observe that $\frac{1}{n}\left[\ell_n(\theta \mid S^1) - \ell_n(\theta_0 \mid S^1)\right] = g_n(\theta) - g_n(\theta_0)$. Because $g_n(\theta_0) \xrightarrow{a.s.} g_\infty(\theta_0) = 0$, the limit supremum of the difference is identical to the limit supremum of $g_n(\theta)$. Therefore

$$\limsup_{n\to\infty} \sup_{\theta\in\Theta\setminus B_\delta(\theta_0)} \frac{1}{n}\left[\ell_n(\theta \mid S^1) - \ell_n(\theta_0 \mid S^1)\right] < 0,$$

with $P_{m(S,\theta_0)}$-probability one, completing the proof.

Since all these conditions are fulfilled, then by Theorem 7 in Kass et al. [1990] the sequence of log-likelihood functions is Laplace regular with $P_{m(S,\theta_0)}-$probability one, for any $\theta_0 \in \Theta$. □

Lemma A.4. *Let $\mathcal{D}_{obs} := (X_i)_{1 \leq i \leq n}$ be n i.i.d. observational samples generated by (1), and let $\mathcal{D}_{int} := (Y_j)_{1 \leq j \leq m}$ be m i.i.d. interventional samples drawn from the post-intervention distribution induced by an intervention $do(Y(2) = y)$. The maximum likelihood estimators for each model are*

$$\hat{\theta}^1 := [\hat{\theta} \mid S^1] = \left[\frac{S_{12}^X + S_{12}^Y}{S_2^X + S_2^Y}, \frac{(S_1^X + S_1^Y)(S_2^X + S_2^Y) - (S_{12}^X + S_{12}^Y)^2}{(n+m)(S_2^X + S_2^Y)}, \frac{S_2^X}{n}\right]^\intercal, \tag{26}$$

$$\hat{\theta}^2 := [\hat{\theta} \mid S^2] = \left[\frac{S_{12}^X}{S_1^X}, \frac{S_1^X + S_1^Y}{n+m}, \frac{S_1^X S_2^X - S_{12}^{X2}}{nS_1^X}\right]^\intercal, \tag{27}$$

$$\hat{\theta}^3 := [\hat{\theta} \mid S^3] = \left[0, \frac{S_1^X + S_1^Y}{n+m}, \frac{S_2^X}{n}\right]^\intercal, \tag{28}$$

where

$$S_{12}^X := \sum_{i=1}^n X_i(1)X_i(2), \quad S_1^X := \sum_{i=1}^n X_i(1)^2, \quad S_2^X := \sum_{i=1}^n X_i(2)^2, \tag{29}$$

$$S_{12}^Y := y\sum_{i=1}^m Y_i(1), \quad S_1^Y := \sum_{i=1}^n Y_i(1)^2, \quad S_2^Y := my^2. \tag{30}$$

Proof. First, we consider the S^1 model. The log-likelihood for this case is

$$\begin{aligned}
\ell_{n,m}(\theta \mid S^1) :&= \sum_{i=1}^n \log f_{\text{obs}}(X_i \mid \theta, S^1) + \sum_{j=1}^m \log f_{\text{int}}(Y_j \mid \theta, S^1) \\
&= \sum_{i=1}^n \Big[\log \phi_{\tau_1^2}(X_i(1) - wX_i(2))^2 + \log \phi_{\tau_2^2}(X_i(2))\Big] + \sum_{j=1}^m \log_{\tau_1^2}(Y_i(1) - wy) \\
&= -\Big(n + \frac{m}{2}\Big)\log 2\pi - \frac{n+m}{2}\log \tau_1^2 - \frac{n}{2}\log \tau_2^2 - \frac{S_2^X}{2\tau_2^2} \\
&\quad - \frac{(S_1^X + S_1^Y) - 2w(S_{12}^X + S_{12}^Y) + w^2(S_2^Y + S_2^X)}{2\tau_1^2},
\end{aligned} \tag{31}$$

where f_{obs} and f_{int} are the densities of the observational and interventional samples, respectively. The maximum likelihood estimates are obtained by solving the first-order conditions $\nabla_\theta \ell_{n,m}(\theta \mid S^1) = 0$. Differentiating with respect to w, we find:

$$\frac{\partial \ell_{n,m}(\theta \mid S^1)}{\partial w} = -\frac{1}{2\tau_1^2}\Big[-2(S_{12}^X + S_{12}^Y) + 2w(S_2^X + S_2^Y)\Big] = 0 \implies \hat{w} = \frac{S_{12}^X + S_{12}^Y}{S_2^X + S_2^Y}.$$

Substituting $\hat{w}$ and differentiating with respect to the variance parameters yields the variance estimators. Moreover, one can easily check that $\ell_{n,m}(\theta \mid S^1)$ is strictly concave in θ by showing that the Hessian is negative definite. Therefore, the maximum is unique. The MLEs for models S^2 and S^3 follow analogously. $\square$

Lemma A.5. *Let $\{n_N\}_{N\geq 1}$ and $\{m_N\}_{N\geq 1}$ be two increasing sequences such that $N = n_N + m_N$, and define $\eta_N := \frac{n_N}{N}$, with $\lim_{N\to\infty} \eta_N = \eta \in (0,1)$. Let $\mathcal{D}_{obs} := (X_i)_{1\leq i\leq n_N}$ be n_N i.i.d. observational samples generated by (1), and let $\mathcal{D}_{int} := (Y_j)_{1\leq j\leq m_N}$ be m_N i.i.d. interventional samples drawn from the post-intervention distribution induced by an intervention $do(Y(2) = y)$. Then, as $N\to\infty$, with $\bar{\eta} = 1-\eta$*

$$\frac{S_2^X}{n_N} \xrightarrow{a.s.} \mathbb{E}[X(2)^2], \tag{32}$$

$$\frac{S_{12}^X + S_{12}^Y}{S_2^X + S_2^Y} \xrightarrow{a.s.} \frac{\eta\mathbb{E}[X(1)X(2)] + \bar{\eta}y\mathbb{E}[Y(1)]}{\eta\mathbb{E}[X(2)^2] + \bar{\eta}y^2}, \tag{33}$$

$$\frac{(S_2^X + S_2^Y)(S_1^X + S_1^Y) - (S_{12}^X + S_{12}^Y)^2}{N(S_2^Y + S_2^X)} \xrightarrow{a.s.} \frac{C}{\eta\mathbb{E}[X(2)^2] + \bar{\eta}y^2}, \tag{34}$$

where S_{12}^X, S_{12}^Y, S_1^Y and S_2^Y are defined in (30) and

$$C = (\eta\mathbb{E}[X(1)^2] + \bar{\eta}\mathbb{E}[Y(1)^2])(\eta\mathbb{E}[X(2)^2] + \bar{\eta}y^2) - (\eta\mathbb{E}[X(1)X(2)] + \bar{\eta}y\mathbb{E}[Y(1)])^2.$$

Proof. Since the second moments are finite, $\mathbb{E}[X(1)^2] < \infty$, $\mathbb{E}[X(2)^2] < \infty$ and $\mathbb{E}[Y(1)^2] < \infty$, then, by the Cauchy-Schwarz inequality, $\mathbb{E}[|X(1)X(2)|] \leq \sqrt{\mathbb{E}[X(1)^2]\mathbb{E}[X(2)^2]} < \infty$ and $\mathbb{E}[|Y(1)|] < \infty$.

Applying the strong law of large numbers to $S_2^X = \sum_{i=1}^{n_N} X_i(2)^2$, we obtain almost sure convergence, $\frac{S_1^X}{n_N} \xrightarrow{a.s.} \mathbb{E}[X(1)^2]$. Next, for the second estimator we have

$$\frac{S_{12}^X + S_{12}^Y}{S_2^X + S_2^Y} = \frac{\eta_N \frac{1}{n_N}\sum_{i=1}^{n_N} X_i(1)X_i(2) + \bar{\eta}_N \sum_{j=1}^{m_N} yY_j(1)}{\eta_N \frac{1}{n_N}\sum_{i=1}^{n_N} X_i(2)^2 + \bar{\eta}_N y^2}.$$

Using the strong law of large numbers, the continuous mapping theorem, and the fact that $\eta_N \to \eta \in (0,1)$, the required result follows. The result for the third estimator is obtained by defining the continuous function $g(x,y,z) = \frac{xy-z^2}{y}$ and following the steps taken in the proof of Lemma A.2. □

Lemma A.6. *Let $B_\delta(\theta)$ denote the open ball of radius δ centered at $\theta \in \Theta := \mathbb{R}\setminus\{0\}\times\mathbb{R}_+^2$. Let $\{n_N\}_{N\geq 1}$ and $\{m_N\}_{N\geq 1}$ be two increasing sequences such that $N = n_N + m_N$, and define $\eta_N := \frac{n_N}{N}$, with $\lim_{N\to\infty}\eta_N = \eta \in (0,1)$. For any $S \in \mathcal{S}$ and $\theta \in \Theta$, let $\mathcal{D}_{obs} := (X_i)_{1\leq i\leq n_N}$ be n_N i.i.d. observational samples generated by (1) with density $f_{obs}(x \mid S,\theta)$, and let $\mathcal{D}_{int} := (Y_j)_{1\leq j\leq m_N}$ be m_N i.i.d. interventional samples drawn from the post-intervention distribution induced by an intervention $do(Y(2) = y)$ with density $f_{int}(x \mid S,\theta)$. The following four conditions hold*

(i) for all $z_1^N = ((x_i)_{1\leq i\leq n_N}, (y_j)_{1\leq j\leq m_N})$ with $x_i \in \mathbb{R}^2$ and $\theta\in\Theta$, $\prod_{i=1}^{n_N} f_{obs}(x_i \mid S,\theta)\prod_{j=1}^{m_N} f_{int}(y_j \mid S,\theta) > 0$. Moreover, for all z_1^N, $\ell_{n_N,m_N}(\theta \mid S) = \sum_{i=1}^{n_N}\log f_{obs}(x_i \mid S,\theta) + \sum_{j=1}^{m_N}\log f_{int}(y_j \mid S,\theta)$ is six times continuously differentiable in θ;

(ii) for any $\theta_0 \in \Theta$ there exist $\epsilon > 0$ and $M < \infty$ such that $B_\epsilon(\theta_0) \subseteq \Theta$ and for $1 \leq j_1,\dots,j_d \leq m$ with $d \leq 6$,

$$\limsup_{N\to\infty} \sup\{N^{-1}|\partial_{j_1}\dots\partial_{j_d}\ell_{n_N,m_N}(\theta \mid S)| : \theta \in B_\epsilon(\theta_0)\} < M,$$

with $P_{m(S,\theta_0)}$-probability one;

(iii) for any $\theta_0 \in \Theta$ and some $\epsilon > 0$,

$$\limsup_{N\to\infty} \sup\{\det(N^{-1}\nabla_\theta^2 \ell_{n_N,m_N}(\theta \mid S)) : \theta \in B_\epsilon(\theta_0)\} < 0$$

with $P_{m(S,\theta_0)}$-probability one;

(iv) for any $\theta_0 \in \Theta$ and any $\delta > 0$,

$$\limsup_{N\to\infty} \sup\{N^{-1}[\ell_{n_N,m_N}(\theta \mid S) - \ell_{n_N,m_N}(\theta_0 \mid S)] : \theta \in \Theta\setminus B_\delta(\theta_0)\} < 0$$

with $P_{m(S,\theta_0)}$-probability one.

Hence, the models are Laplace regular as per Kass et al. [1990].

Proof. The proof follows the same steps as in the proof of Lemma A.3. For simplicity, we assume $S = S^1$. The remaining cases follow analogously. Using the expression for the ML estimator, $\hat{\theta}^1$, derived in Lemma A.4 and (31), the log-likelihood is

$$\begin{aligned}\ell_{n_N,m_N}(\theta \mid S^1) &= -\left(n_N + \frac{m_N}{2}\right)\log 2\pi - \frac{n_N}{2}\log(\tau_1^2\tau_2^2) - \frac{m_N}{2}\log\tau_1^2 - \frac{(n_N\hat{\tau}_2^2 + m_N y^2)(w-\hat{w})^2}{2\tau_1^2} \\ &\qquad - \frac{(n_N+m_N)\hat{\tau}_1^2}{2\tau_1^2} - \frac{n_N\hat{\tau}_2^2}{2\tau_2^2} \\ &= -N\Bigg\{\left(\eta_N + \frac{\bar{\eta}_N}{2}\right)\log 2\pi - \frac{\eta_N}{2}\log(\tau_1^2\tau_2^2) - \frac{\bar{\eta}_N}{2}\log\tau_1^2 \\ &\qquad - \frac{(\eta_N\hat{\tau}_2^2 + \bar{\eta}_N y^2)(w-\hat{w})^2}{2\tau_1^2} - \frac{\hat{\tau}_1^2}{2\tau_1^2} - \frac{\eta_N\hat{\tau}_2^2}{2\tau_2^2}\Bigg\},\end{aligned} \tag{35}$$

where $\bar{\eta}_N = 1 - \eta_N$.

(i)-(ii) Similarly to the proof of Lemma A.3, the first two conditions are easily verifiable; i.e. $\ell_{n_N,m_N}(\theta \mid S^1)$ is 6-times differentiable and $f(z_1^N \mid S^1, \theta) > 0$ for all $\mathcal{D}_{\text{mix}} = \mathcal{D}_{\text{obs}} \cup \mathcal{D}_{\text{int}}$ and $\theta \in \Theta$. Moreover, $\ell_{n_N,m_N}(\theta \mid S^1)$ is composed of terms that are either quadratic polynomials in w and rational functions of the variance parameters τ_1^2 and τ_2^2 and hence the partial derivatives are bounded in a vicinity of θ_0.

(iii) The Hessian is

$$H(\theta) := H(w, \tau_1^2, \tau_2^2) = N\begin{bmatrix} -\frac{(\eta_N\hat{\tau}_2^2+\bar{\eta}_N y^2)}{\tau_1^2} & \frac{(\eta_N\hat{\tau}_2^2+\bar{\eta}_N y^2)(w-\hat{w})}{(\tau_1^2)^2} & 0 \\ \frac{(\eta_N\hat{\tau}_2^2+\bar{\eta}_N y^2)(w-\hat{w})}{(\tau_1^2)^2} & \frac{1}{2(\tau_1^2)^2} - \frac{\left[\hat{\tau}_1^2+(\eta_N\hat{\tau}_2^2+\bar{\eta}_N y^2)(w-\hat{w})^2\right]}{(\tau_1^2)^3} & 0 \\ 0 & 0 & \frac{\eta_N}{2(\tau_2^2)^2} - \frac{\eta_N\hat{\tau}_2^2}{(\tau_2^2)^3}\end{bmatrix} \tag{36}$$

with the determinant of the normalized Hessian being

$$h_N(\theta) := \det\left(\frac{1}{N}H(\theta)\right) = -\frac{\eta_N(\eta_N\hat{\tau}_2^2 + \bar{\eta}_N y^2)}{4\tau_1^6\tau_2^4}\left(1 - \frac{2\hat{\tau}_2^2}{\tau_2^2}\right)\left(1 - \frac{2\hat{\tau}_1^2}{\tau_1^2}\right).$$

Let $\theta_0 := \left(w_0, \tau_{1,0}^2, \tau_{2,0}^2\right) \in \Theta$ be the true generating parameter. Furthermore, we note that both estimators $\hat{\tau}_1{}^2$ and $\hat{\tau}_2{}^2$ are consistent per (26) and Lemma A.5,

$$\hat{\tau}_1^2 \xrightarrow{a.s.} \tau_{1,0}^2 \quad \text{and} \quad \hat{\tau}_2^2 \xrightarrow{a.s.} \tau_{2,0}^2,$$

because $\mathbb{E}[X(1)^2] = w_0^2\tau_{2,0}^2 + \tau_{1,0}^2$, $\mathbb{E}[X(2)^2] = \tau_{2,0}^2$, $\mathbb{E}[Y(1)^2] = w_0^2 y^2 + \tau_{1,0}^2$, $\mathbb{E}[X(1)X(2)] = w_0\tau_{2,0}^2$ and $\mathbb{E}[Y(1)Y(2)] = w_0 y^2$. Since the estimators are consistent and the function $h_N(\theta)$ resembles the definition of $h_n(\theta)$ from (21), the desired result follows from the same arguments presented in the proof of condition (iii) in Lemma A.3.

(iv) Using the expansion in (35), the normalized difference between the log-likelihood at a parameter $\theta = [w, \tau_1^2, \tau_2^2]^\top$ and at the MLE is given by

$$\begin{aligned} g_N(\theta) &:= \frac{1}{N}\left[\ell_{n_N,m_N}(\theta \mid S^1) - \ell_{n_N,m_N}(\hat{\theta}^1 \mid S^1)\right] \\ &= -\frac{1}{2}\left[\frac{(\eta_N\hat{\tau}_2^2 + \bar{\eta}_N y^2)(w-\hat{w})^2}{\tau_1^2} + \phi\left(\frac{\hat{\tau}_1^2}{\tau_1^2}\right) + \eta_N\phi\left(\frac{\hat{\tau}_2^2}{\tau_2^2}\right)\right], \end{aligned}$$

where we define $\phi(u) = u - 1 - \log u$ for $u > 0$. The function $g_N(\theta)$ is structurally analogous to $g_n(\theta)$ defined in (22), differing only by the existence of the η_N for which $\eta_N \to \eta$ as $N \to \infty$. Consequently, the result follows from the arguments presented in the proof of condition (iv) in Lemma A.3.

$\square$